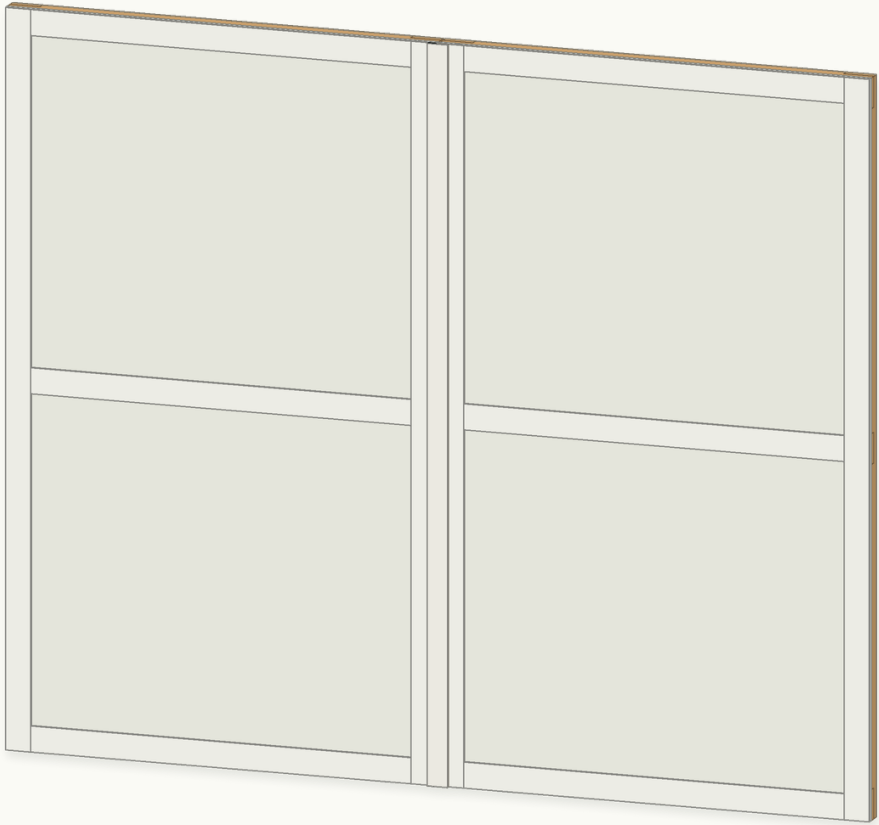


Garage door assembly

An illustrated guide for rebuilding your existing outward-opening pair.



SELECT MATERIALS IN IMPERIAL; CUT IN cm

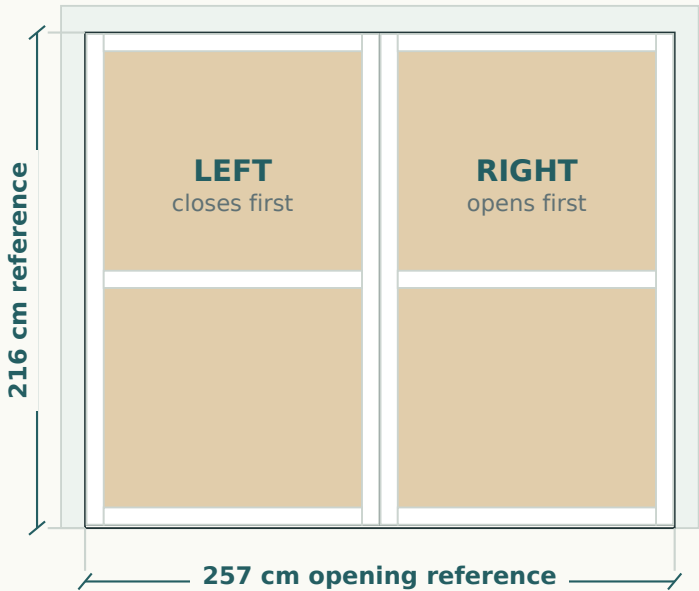
Outside plywood	Flat timber core	Inside plywood	White trim
15/32 in thick	2×4; actual 1½ × 3½ in	1/5 in thick	Model: 0.236 in thick

Use imperial labels for tools, fasteners and stock. Cuts and workshop marks are in cm. Keep sound existing framing; replacement half-laps apply only where needed.

Find the next action	Pages
Measure, gather tools and identify parts	2-5
Prepare frame, backing, plywood and insulation	6-13
Fit trim, finish, hang and seal	14-19
Record checks; workshop details; source notes	20-27

Before you cut

Record the existing door sizes and opening. All sizes in this manual remain field-fit references.



MEASURE AND WRITE

Opening widths: top / middle / bottom

Opening heights: left / center / right

Both opening diagonals

Existing LEFT: width / height / depth

Existing RIGHT: width / height / depth

Floor high point through the full swing

TOOLS AND WORKSPACE

Tape measure, square, straightedge/level, pencil, clamps, flat supports, drill/driver and bits. A guided saw for plywood; suitable joint-cutting tools only for replacement frame members. Add sanding/finishing supplies, eye/hearing/dust protection, and a helper with suitable lifting/support equipment.

CONFIRM THESE BEFORE ASSEMBLY

- ☐ **Material condition:** dry, flat timber and sound joints. Record existing braces before opening the leaf; keep sound reinforcement.
- ☐ **Products:** verify the outside plywood is suitable for permanent exterior use; select the foam and its required protective covering. The 1/5 in liner has no established fire/thermal-barrier rating. [1]
- ☐ **Hardware:** obtain the actual hinge, latch and bolt templates, including fastener requirements and access to any nuts or backing plates.
- ☐ **Bonding and finish:** choose compatible products before priming. Coatings on a glue surface may prevent the required bond. [3, 4]

DIMENSION RULE

If the measured opening, retained frame or required covering differs, revise the cut sizes and weight calculation together. Equal leaves are an assumption in this model.

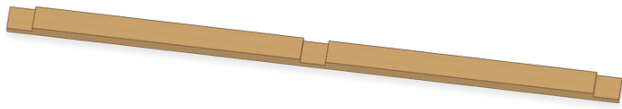
1. Frame and plywood

QTY = total for BOTH doors

Stock in inches / feet. Cut lengths in cm; keep stock widths where stated.

F1 Frame uprights × 4

2 × 4 timber; actual 1-1/2 × 3-1/2 in



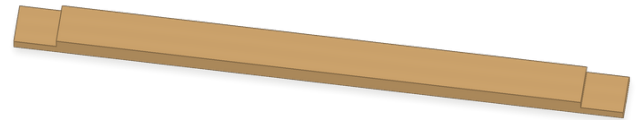
Cut length: 213.865 cm

CUT ONLY IF REPLACING

Keep stock width. Cut length and **3 half-laps**. Keep sound existing members.

F2 Frame cross rails × 6

2 × 4 timber; actual 1-1/2 × 3-1/2 in



Cut length: 127.565 cm

CUT ONLY IF REPLACING

Keep stock width. Cut **full leaf width** and 2 end half-laps; retain sound rails.

P1 Outside main panels × 2

15/32 in exterior plywood; 4 ft wide



Cut length: 213.865 cm

YES - CUT HEIGHT

Shorten to the shown length. Keep the actual usable stock width.

P2 Outside infill strips × 2

15/32 in matching exterior plywood



Cut: 213.865 × 5.325 cm

YES - CUT TO SIZE

Cut width and length unless your existing strips fit. Support on timber.

P3 Inside main panels × 2

1/5 in inner plywood; 4 ft wide



Cut length: 213.865 cm

YES - CUT HEIGHT

Shorten to the shown length. Keep the actual usable stock width.

P4 Inside infill strips × 2

1/5 in matching inner plywood



Cut: 213.865 × 5.325 cm

YES - CUT TO SIZE

Cut width and length unless your existing strips fit. Support on timber.

Half-lap cuts: 8.89 cm long × 1.905 cm deep for the stated stock. Mark from actual timber. Pictures use separate scales. Decimal precision is not a cutting tolerance.

2. Insulation, trim and joint

QTY = total for BOTH doors

Stock thickness in inches. Cut length × width in cm; seal is length only.

I1 Insulation inserts × 4

1-1/2 in rigid foam



Cut: 109.485 × 93.298 cm

YES - CUT RECTANGLES

Cut length and width. Fit flush in the actual frame cavities.

T1 Vertical white trim × 4

Model thickness: approx. 0.236 in*



Cut: 213.865 × 7.5 cm

YES - CUT TO SIZE

Cut length and width. One at each vertical edge of each door.

T2 Horizontal white trim × 6

Model thickness: approx. 0.236 in*



Cut: 112.565 × 7.5 cm

YES - CUT TO SIZE

Cut length and width. Three between the vertical trims on each door.

C1 White center cover × 1

Model thickness: approx. 0.236 in*



Cut: 213.865 × 6 cm

YES - CUT AND FIT

Cut length and width; fit both ends. Attach to the **RIGHT door only**.

C2 Center-cover spacer × 1

Model thickness: approx. 0.118 in*



Cut: 213.865 × 3 cm

YES - CUT AND FIT

Cut length and width. Thickness follows the selected gasket; **RIGHT only**.

S1 Center rubber seal × 1

Model section: 0.866 × 0.118 in**



Cut length: 213.865 cm

CUT TO LENGTH ONLY

Choose the seal profile, then cut its length. **Do not cut its width** to this model.

*Model thicknesses are not exactly 1/4 in or 1/8 in; measure actual stock. **Seal section is a compressed model, not a purchase specification. Confirm all cuts on the actual door.

3. Hardware and edge seals

QTY = total for BOTH doors

Imperial product sizes. Schematic examples; hardware remains to be selected.

H1 Load-bearing hinges **× 6**



Size in inches: to select

NO METAL CUTTING

Three per leaf shown. Confirm rating and timber backing before mounting.

H2 Door pulls **× 2**



Size in inches: to select

NO PART CUTTING

Use factory parts. Drill mounting holes to the selected template.

H3 Latch arrangement **× 1**



Size in inches: to select

NO PART CUTTING

Right active door. Fix to solid timber; fit to the selected template.

H4 Left-door bolts **× 2**

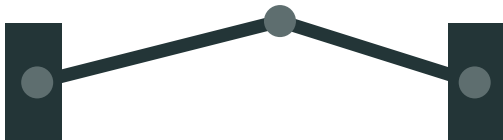


Size in inches: to select

NO PART CUTTING

One top and one bottom on the left leaf. Fit keepers to sound supports.

H5 Hold-open devices **× 2**



Size in inches: to select

NO PART CUTTING

One per leaf. Select and position for the actual outward swing.

H6 Perimeter seals **× 5 runs**



Cuts: head 257; sides 216 cm

YES - CUT TO FIT

Sides: **2 runs**. Bottoms: **2 × 127.565 cm** reference. Select profiles; measure final cuts.

Seal lengths are planning runs, not final cuts. Extra blocking, screws, adhesives and coatings depend on the actual frame and selected products; their quantities are not fixed.

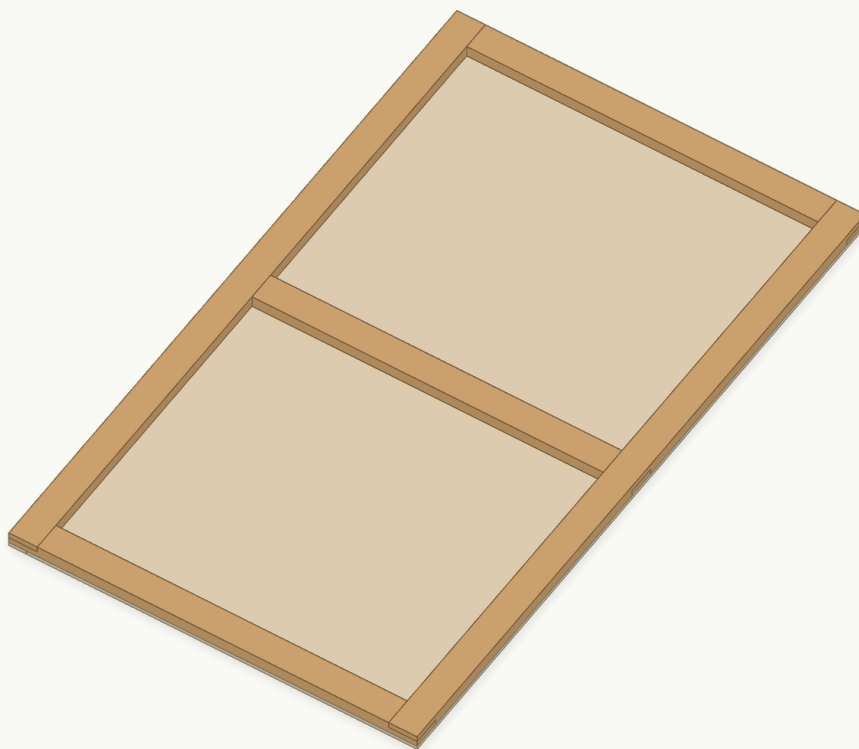
01 Open and label the existing leaf

Work on one leaf at a time. This drawing shows the reference frame, not an inspection of your existing door.

LEFT / RIGHT

OUTSIDE / INSIDE

TOP / BOTTOM



- 1 Support the leaf securely before releasing its hinges.** Mark its handing, top, hinge edge and outside face. Photograph hinge positions and measure the original gaps.
- 2** Lower it onto flat, padded supports. Remove the skins carefully enough to preserve sound framing, joints and braces.
- 3** Inspect for decay, splits, loose joints, twist and loose fixings. Mark the members that actually need repair; keep the others.

CHECK BEFORE CONTINUING

You have recorded the actual frame layout and hardware locations. The leaf is supported flat, and every proposed repair is identified.

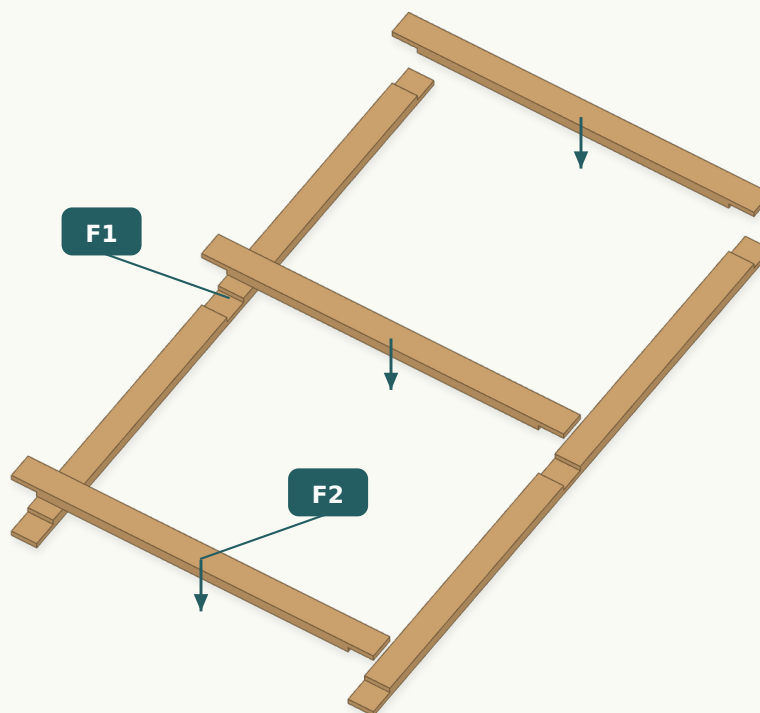
02 Repair and square the frame

For each reference leaf: two uprights and three full-width cross rails. Replacement cuts are conditional.

F1 × 2 per door

F2 × 3 per door

Keep sound existing joints



- 1 Lay out the retained or replacement members on a flat bench. For new reference members, use the complementary half-laps on the next page.
- 2 Clamp to **127.565 cm wide × 213.865 cm high**. Compare both diagonals and adjust the frame until it is square and flat.
- 3 Complete the confirmed joint fastening/bonding arrangement. Keep the frame square while it sets; retain or provide the required racking reinforcement.

FRAME CHECK

Nominal rectangle diagonal: 249.020 cm in both directions. Geometry alone does not define joint strength: confirm the joint fixings and racking resistance for the actual timber and leaf load. [5]

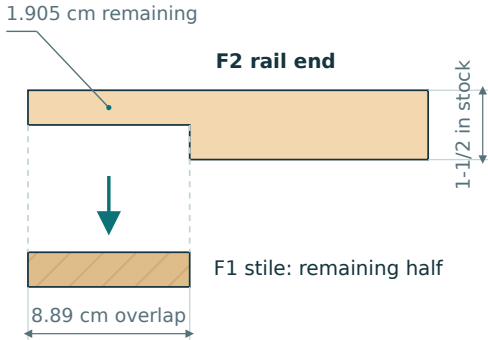
02A Half-lap cuts for replacements

Use this page only when rebuilding frame members. Cut half the actual timber thickness, not blindly to a nominal value.

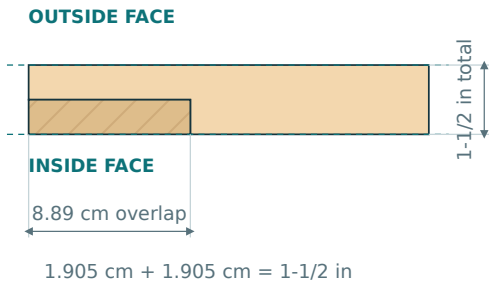
HALF-LAP: FIT THE RAIL INTO THE STILE

Replacement members only. Keep sound existing frame joints.

1 BRING THE MATCHING CUTS TOGETHER



2 BOTH FACES FINISH FLUSH



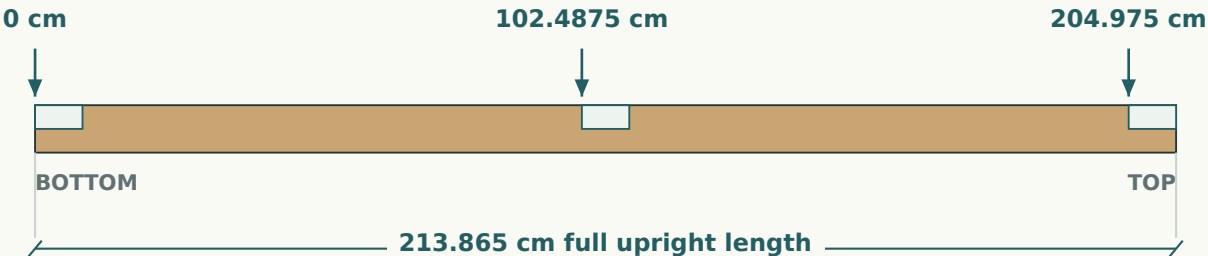
F2 full rail blank: 127.565 cm long.

Cut complementary laps to half the actual timber thickness; dry-fit flat before fixing.

Section through joint

Stock: imperial • Cuts / marks: cm • V4 model reference • Not full size

F1: NOTCH LOWER EDGES FROM THE LEAF BOTTOM



Cut orientation: remove the outside half of F1 at the crossings and the inside half of F2 at its ends. Each notch is **8.89 cm long × 1.905 cm deep** in actual 1½ in thick timber.

Fit: trial-fit one lap before repeating cuts. Broad mating faces meet fully, both door faces finish flush, and the total frame depth remains **1½ in**.

CUTTING CHECK

The F2 rail blank is the full 127.565 cm leaf width. Its ends overlap the uprights; do not shorten it to the 109.785 cm clear space between them.

03 Prepare hardware backing

Select the actual hardware first. Every loaded fixing needs sound wood behind it.

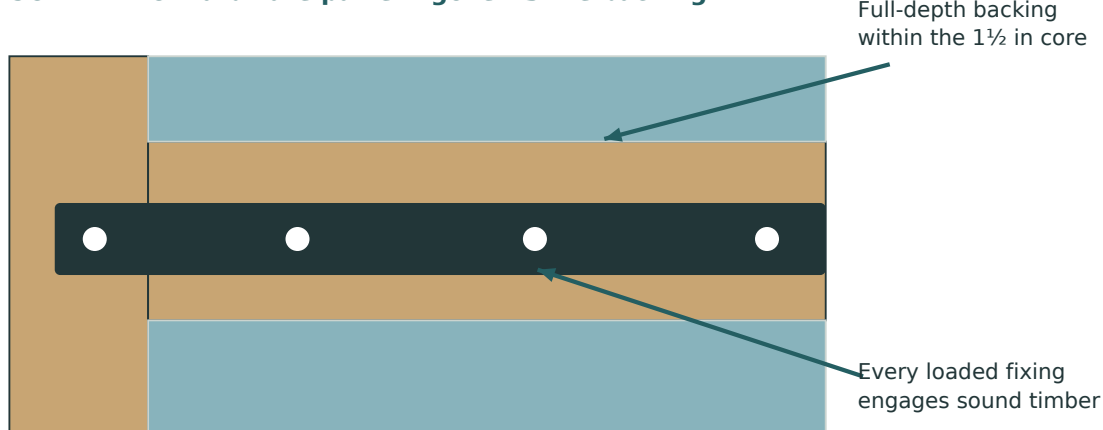
H1 hinges

H3 latch

H4 bolts

Backing: field-sized

SCHEMATIC: hardware pattern governs the backing



No hole positions or extra-blocking cut sizes are specified by this sketch.

- 1 Place the real hinge, latch and bolt templates on the frame. Mark the hole centers, edge clearances and jamb/keeper locations.
- 2 Align the hardware with suitable rails/stiles or add securely connected full-depth blocking to its exact pattern. Update the foam cuts and weight allowance.
- 3 Complete access-dependent fasteners, nuts or backing plates before closing the skins. Record the selected fixings on page 20.

BEFORE COVERING THE FRAME

All load paths reach solid door framing and a sound structural jamb. Thin plywood and foam are not the hinge anchorage. The illustration adds example backing; it is not part of the base cut count.

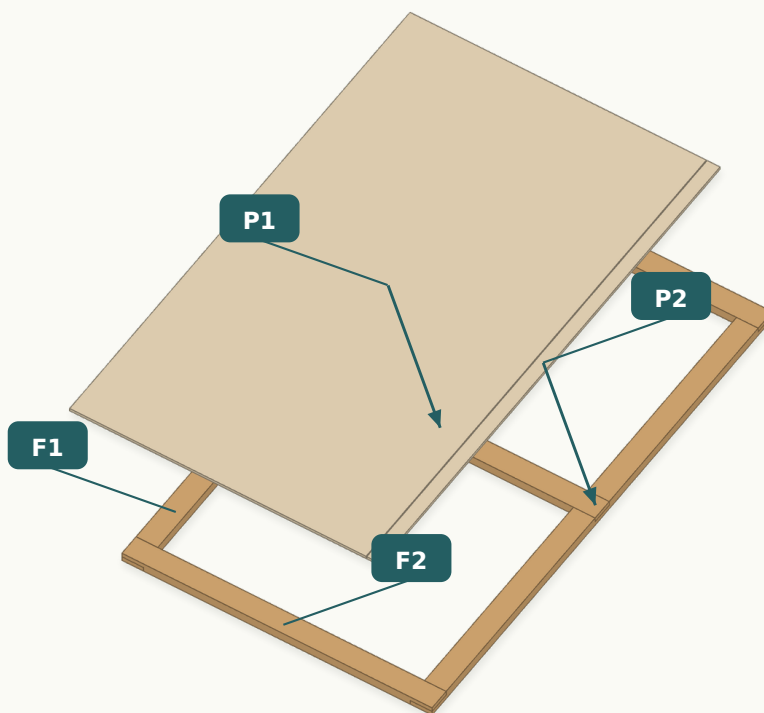
04 Attach the outside plywood

Keep the frame flat and square. Mirror the two leaves so their narrow strips meet at the center.

P1 × 1 per door

P2 × 1 per door

Outside face up



- 1 Dry-fit **P1: 213.865 × 121.92 cm** and the matching **P2 strip: 213.865 × 5.325 cm**. Use 15/32 in outside plywood.
- 2 Put the narrow strip at the meeting edge. Keep the reference **0.32 cm seam** between the pieces, with timber beneath both edges; see the next page. [2]
- 3 Mark the timber lines on the face. Attach the skins with the confirmed wood/substrate bonding and fastening schedule; keep fixings over timber and recheck square.

BEFORE THE FINAL RIP CUT

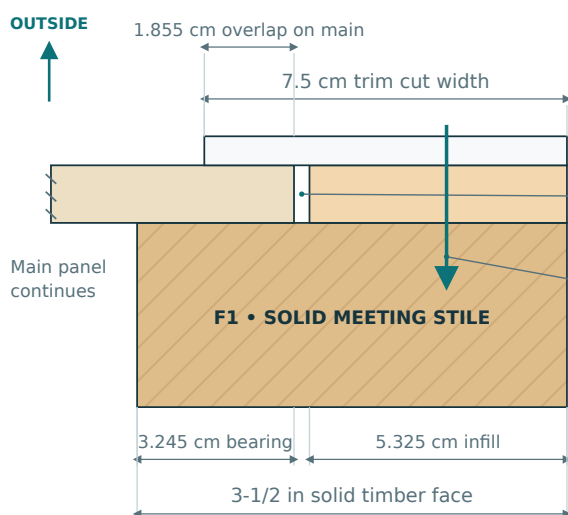
Measure the usable sheet width first. Infill width = measured leaf width - usable main-panel width - selected seam. Protect cut edges, while keeping glue surfaces compatible with the chosen adhesive.

04A Keep the seam supported

The narrow strip makes up the extra door width. The white meeting trim hides the seam after assembly.

PLYWOOD: SUPPORT THE SEAM ON TIMBER

Representative left meeting stile shown. Mirror this detail for the right door.



THE OUTSIDE LAYERS

- ☐ T1 Trim: approx. 0.236 in thick
- ☐ P1 Main plywood: 15/32 in thick
- ☐ P2 Matching 15/32 in infill

0.32 cm movement seam
Keep the gap open.

Fix trim through the infill side into the solid timber.

Do not rigidly glue the trim across both edges of the seam.

Recalculate infill width from the actual usable width of your main plywood sheet.

Stock: imperial • Cuts / marks: cm • V4 model reference • Not full size

Top section

- 1** Keep the seam over the meeting stile **at the location shown**. At the model sizes, the main sheet bears on **3.245 cm** of timber and the strip bears on **5.325 cm**.
- 2** Preserve the **0.32 cm** movement seam. Avoid rigid wood filler or a continuous rigid glue bridge across it. Follow the chosen panel and weather-detail instructions. [2]
- 3** Later, fix the **7.5 cm** meeting trim through the strip side into the stile. Its opposite edge covers the main sheet while allowing the joint to move.

REPEAT ON THE INSIDE

Apply the same supported-seam layout to P3/P4 on the inside. The shown exterior trim is the seam cover; the thin interior lining has no added cover strip in the base part list.

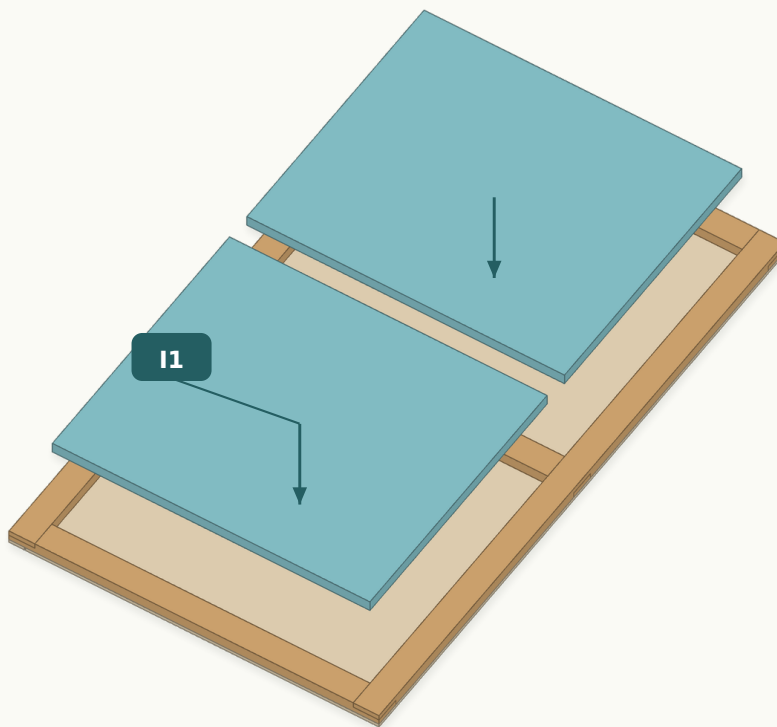
05 Fit the insulation inside

Turn only after the outer-skin bonding system permits handling. Set the exterior face down on padded, flat supports.

11 × 2 per door

Inside face up

1½ in core



- 1 Measure the actual cavities after repairs and hardware backing. Cut each unblocked reference insert to **109.485 × 93.298 cm** from **1½ in** foam; two per leaf.
- 2 Cut and trial-fit each insert without forcing it. Allow for adhesive bedding so the foam finishes flush with, or below, the frame plane and cannot bow the inner skin.
- 3 Use adhesive approved for the exact foam/facing and substrates. Follow its application, retention and full-cure instructions; keep the assembly supported flat. [3]

BEFORE CLOSING THE INSIDE

A straightedge across the frame does not rock on proud foam. Hardware fixings remain accessible as required. Verify the selected foam covering requirements before fitting the 1/5 in lining.

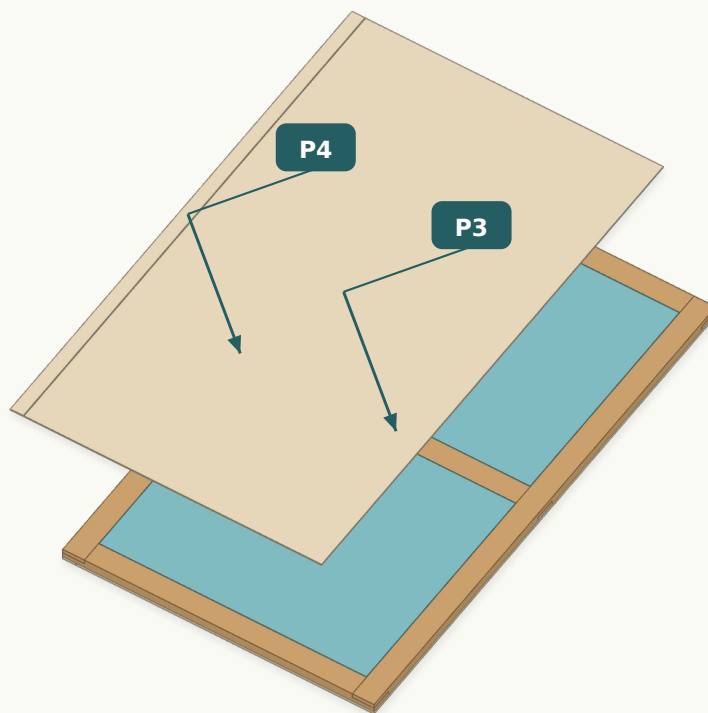
06 Close the inside face

Use the thinner sheet as an interior lining. Keep the narrow strip on the meeting stile.

P3 × 1 per door

P4 × 1 per door

1/5 in lining



- 1 Dry-fit **P3: 213.865 × 121.92 cm** and **P4: 213.865 × 5.325 cm**. Preserve the supported seam just as on the outside.
- 2 Transfer the rail, stile and backing lines to the lining. Attach using the confirmed fixing schedule into wood; seat the heads without crushing or pulling through the thin sheet.
- 3 Support flat for the required adhesive cure. Check the diagonal, face flatness, exposed edges and all hardware access before proceeding.

LINING CHECK

The lining does not become a structural hinge backing, nor a verified protective covering for every foam product, merely because it covers the cavity. Use the selected material instructions.

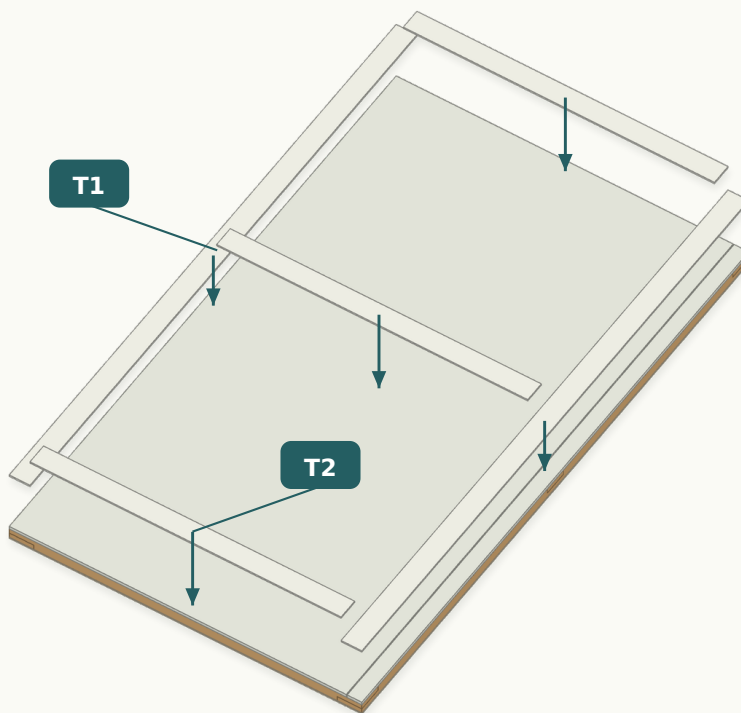
07 Fit the white decorative trim

Dry-fit the pieces on the outside faces of both leaves. Use the same horizontal reference line across the pair.

T1 × 2 per door

T2 × 3 per door

Cut 7.5 cm wide; stock 0.236 in



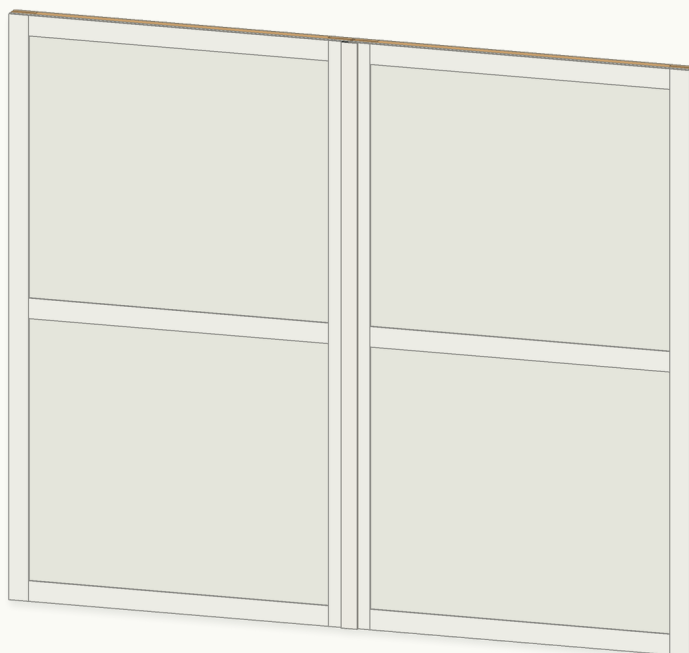
- 1 Place the two full-height **T1 verticals** at the outside edges of each leaf. Fit the three **112.565 cm T2 rails** between them.
- 2 Reference rail lower edges are **0, 103.1825 and 206.365 cm** above the leaf bottom. The middle trim center is **106.9325 cm**. Keep the two doors visually aligned.
- 3 Use the proposed decorative fixing layout on **page 23**: 10 screws per T1, 6 per T2, into timber. The meeting-seam section is on page 22. Ease exposed edges and protect cut ends.

ALIGNMENT CHECK

The trim is cut 7.5 cm wide; the hidden 2×4 frame has a 3½ in face. Their middle centerlines agree; their lower-edge measurements differ. Do not use the trim marks to cut frame notches.

08 Seal and finish all exposed wood

Protect the door before installation, and reseal any later cuts or drilled preparations.



OUTSIDE

INSIDE

TOP + BOTTOM

HINGE + MEETING EDGES

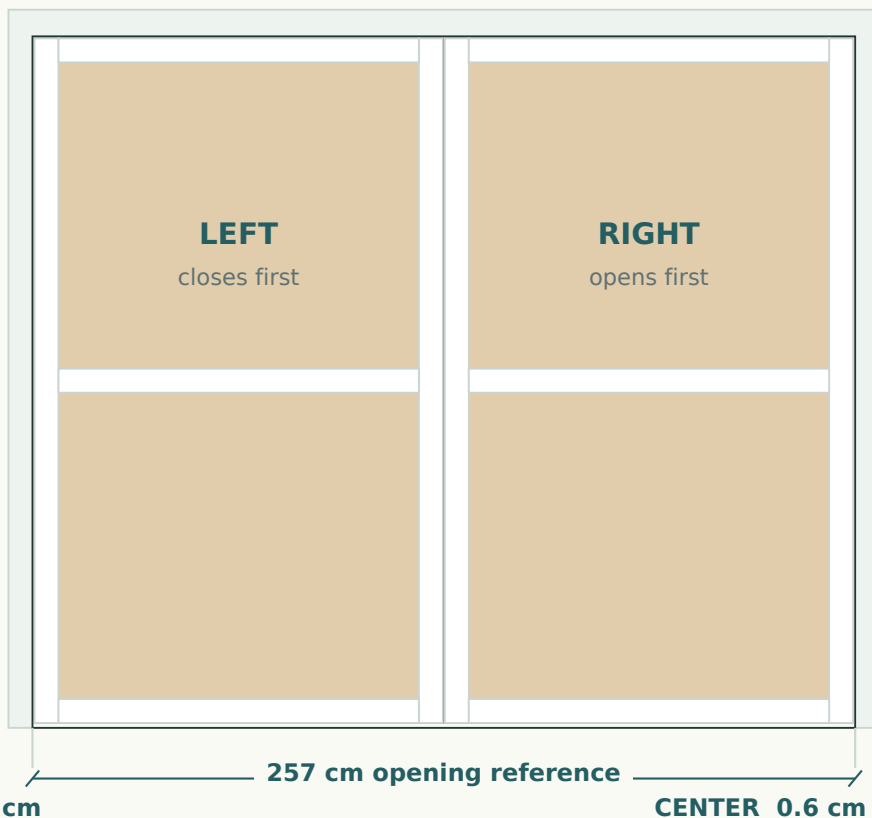
- 1 Sand and remove dust. Ease sharp exposed edges enough for coating coverage, without changing joint bearing or the intended fit.
- 2 Follow the selected exterior finish system. Cover both faces, all four edges, trim ends and cut plywood edges. Preserve any intentionally bare bonding areas until their adhesive operation is complete. [4]
- 3 Let the coating cure as directed. After fitting hardware or trimming the center cover, reseal the new holes/cuts and touch up disturbed areas.

FINISH CHECK

Check the top and bottom edges with the leaf still on supports. Those edges are easy to miss after hanging. Coating counts and cure times come from the selected product, not this drawing.

09 Weigh, check hinges, then hang

Complete the hardware suitability check before a trial hang. Use a helper and secure lifting/support equipment.



- 1 Record each leaf's actual weight with all fitted parts; include the measured weight of anything still to be added. Confirm the hinge system, joint connections, fasteners and jamb for the actual width, weight and exposure.
- 2 Set the **left leaf** on secure temporary supports, then the right. Use the measured gaps and actual hinge template. Keep hinge axes aligned and the leaves level with one another.
- 3 Fasten to the selected specification while keeping the leaf supported. Test the full outward swing over the highest floor point before removing temporary supports.

FIT CHECK

The drawn gaps are starting allowances. Actual hinge geometry, trim depth, floor slope and weatherseal compression control the fit. Do not use the model's ideal hinge pivot as a drilling template.

10 Fit the center cover and seal

The right leaf is active. The cover, spacer and gasket move with it.

C1 cover × 1

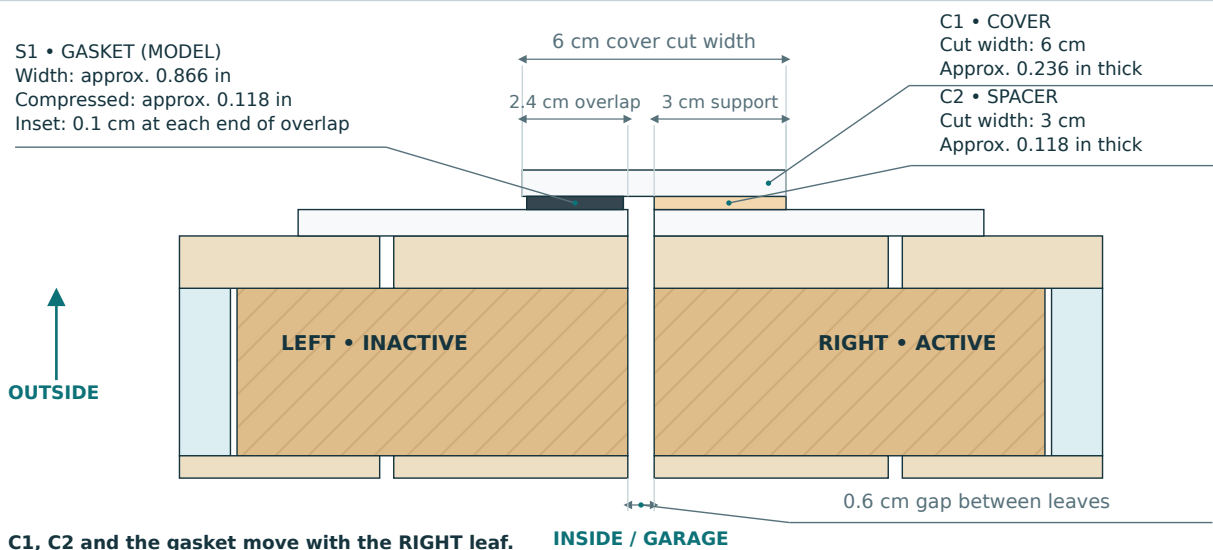
C2 spacer × 1

S1 gasket × 1

RIGHT ONLY

CENTER CONNECTION: THE COVER BELONGS TO THE RIGHT DOOR

Top section through the closed doors. Spacer and gasket thicknesses are working-gap examples.



Stock: imperial • Cuts / marks: cm • V4 model reference • Not full size

Right opens first. Left closes first.

- 1 Hold the left leaf closed and secure it. With both leaves aligned, confirm the center gap and dry-fit the cover across it: **3 cm support on the right + 0.6 cm gap + 2.4 cm overlap on the left.**
- 2 Select the real gasket, then set C2 to its required compressed working gap. The drawing's **0.118 in spacer and compressed gasket** are examples, not the seal's uncompressed purchase thickness.
- 3 Cut cover, spacer and seal ends to the fitted head/bottom details. Fix C1/C2 into the **right meeting timber only**. Fit S1 under the left overlap so it compresses against the left leaf.

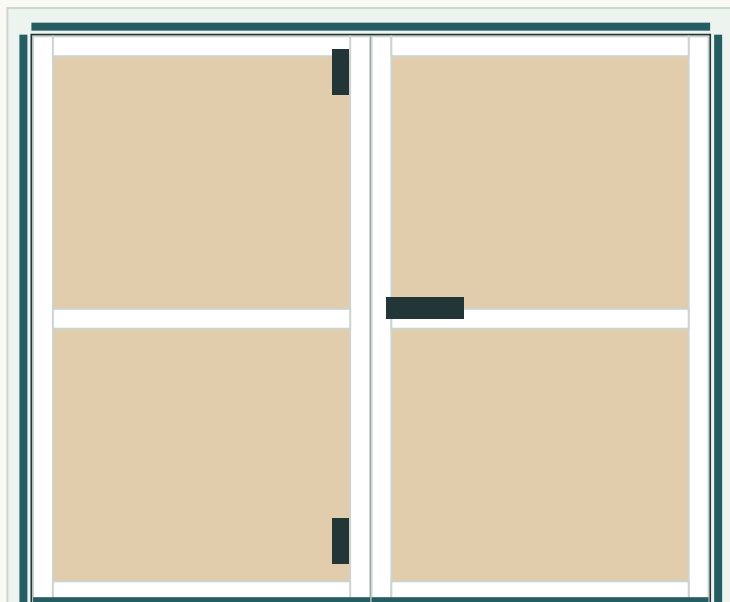
CENTER-JOINT CHECK

The cover releases when the right leaf opens. No screw or rigid glue joint crosses into the left leaf. Check even compression and easy latch action along the full height.

11 Fit the latch, bolts and seals

Use the selected hardware and weatherseal templates. Fit each seal to the actual installed gap.

FIT TO SOUND BACKING



LEFT

Two bolts: top and bottom.

RIGHT

Active latch and center cover.

BOTH

Pulls and hold-open devices.

PERIMETER

Head + two sides + two bottoms.

Schematic positions only; use the manufacturer's mounting templates.

- 1 Install the two left-leaf bolts and their keepers into suitable door framing and supports. Fit the right-leaf latch, pulls and both hold-open devices.
- 2 Fit the head and jamb seals to sound stops. Fit the two bottom seals to the actual threshold/floor condition without using them to support door weight.
- 3 Join the center seal to the head/bottom weather details. Check for continuous contact, free movement, water-shedding edges and no trapped lower-edge water.

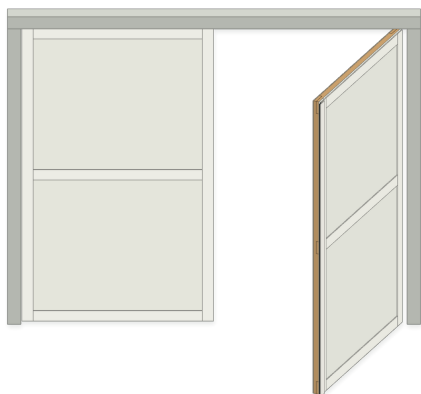
SEAL LENGTHS

Reference runs: head 257 cm; sides 2 × 216 cm; bottoms 2 × 127.565 cm. These are measuring allowances. Different locations may need different profiles; final cut lengths follow the installed detail.

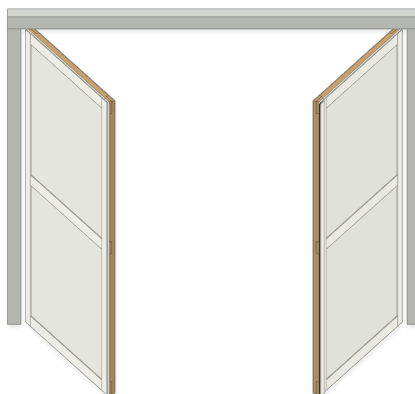
12 Test the opening sequence

Operate slowly first. Support and adjust any leaf that rubs, drops or loads the latch.

1 OPEN RIGHT FIRST



2 RELEASE / OPEN LEFT



OPEN: unlatch and open the right leaf; release the left bolts; open the left. Secure both with their hold-open devices.

CLOSE: close the left leaf and secure its bolts first. Close and latch the right leaf over the center gasket last.

- ☐ Both leaves swing fully without rubbing the floor, jamb, post or one another.
- ☐ All hinges and anchors remain firm; the latch closes without lifting the door.
- ☐ The cover releases freely, the gasket contacts evenly, and the seals do not bind.
- ☐ Finish is complete on edges, ends and later hardware cuts.

AFTER ASSEMBLY

Recheck after initial use and seasonal weather changes. Investigate sag, loose fasteners, damaged finish or deteriorated seals before they worsen.

Complete the assembly record

Use this page with the actual hardware instructions. The manual does not invent a structural screw schedule.

Record	LEFT leaf	RIGHT leaf
Finished width / height, cm	_____	_____
Actual final weight, lb	_____	_____
Hinge model / number	_____	_____
Fastener specification / embedment, in	_____	_____
Jamb material / sound anchorage	_____	_____

FASTENER REACH: CHECK THE MATERIAL STACK

The values below are the layers crossed **before** the screw reaches timber. Add the selected fastener's required timber embedment, then check that its tip and head suit the actual assembly. These are geometry checks, not selected screw lengths.

Fixing location	Layers before timber, inches
Outside skin	$15/32 = 0.46875$
Inside lining	$1/5 = 0.200$
White trim through outside skin	$0.23622 + 0.46875 \approx 0.70497$
Center cover, example spacer, trim and skin	$0.23622 + 0.11811 + 0.23622 + 0.46875 \approx 1.05930$
Hinge or other hardware	Actual hardware + actual intervening layers

- ☐ Frame joints / reinforcement confirmed; access-dependent fixings completed.
- ☐ Actual hinge, latch, bolt and hold-open instructions kept with the project.
- ☐ Both opening directions and all seal transitions checked; later cuts resealed.

Which strip do you cut?

Six different pieces have different jobs. Quantities below cover both doors.

ID	Qty	Cut L × W, cm	Stock thickness, in	Job / cuts
P2	2	213.865 × 5.325	15/32	Outside infill. Rip width; square ends.
P4	2	213.865 × 5.325	1/5	Inside infill. Rip width; square ends.
T1	4	213.865 × 7.5	0.236 model	Vertical trim. Cut length; rip if wider.
T2	6	112.565 × 7.5	0.236 model	Horizontal trim. Square ends between T1.
C1	1	213.865 × 6	0.236 model	RIGHT center cover. Fit final end cuts.
C2	1	213.865 × 3	0.118 example*	RIGHT spacer. Fit thickness and ends.

CUTTING SEQUENCE

- 1

Measure before ripping. Infill width = actual leaf width – usable main-sheet width – joint gap. The model gives **127.565 – 121.92 – 0.32 = 5.325 cm**. Existing strips can be reused if their size, flatness and exterior suitability match.
- 2

Support the sheet and the offcut. Clamp a straight guide to a stable work surface. Put the blade kerf on the waste side of the finished-width mark. Keep the narrow strip supported; use the saw’s guarding and narrow-cut method.
- 3

Make square end cuts. T2 uses simple **90° butt ends** between T1; no 45° miters or half-laps. Dry-fit before the final end trim. Cut one test piece, then repeat with a stop or matched marks.
- 4

Label each piece. Mark its ID, leaf and face on the back. Sand the cut edge and protect it with the chosen finish system, preserving any areas required bare for a later bond. [4]

STOCK AND FIT

*C2 follows the gasket’s compressed working gap; 0.118 in is an example. Trim model: 0.236 in, not exactly ¼ in. Verify actual stock and door dimensions before cutting.

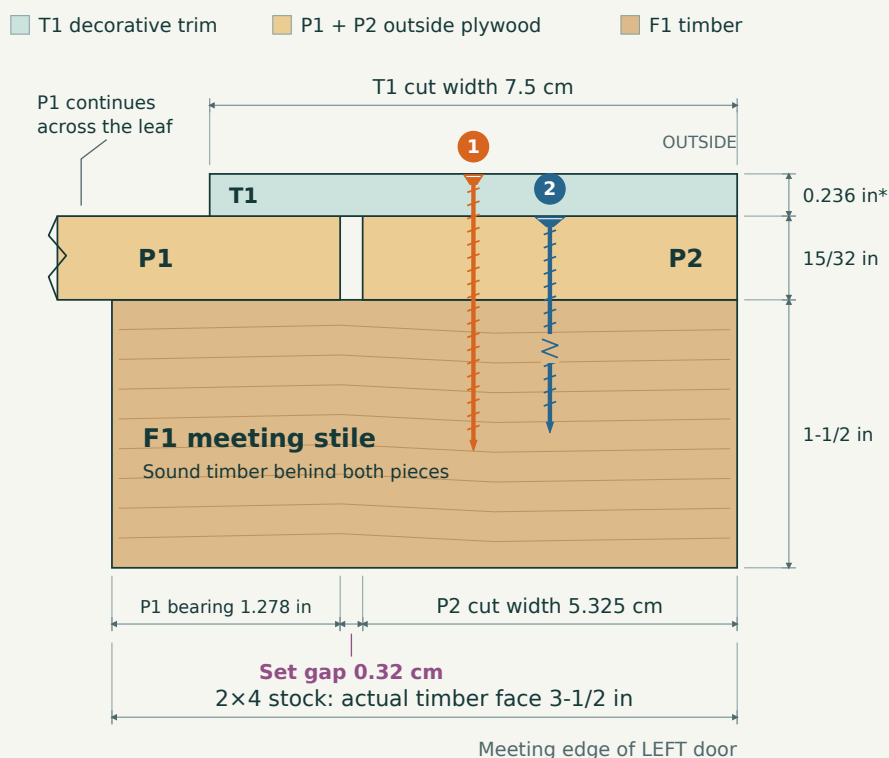
Fix the plywood strip first

P2 fills the skin width. T1 is a separate decorative cover over the supported seam.

GARAGE DOORS / ASSEMBLY DETAIL

How the narrow strips are fixed

LEFT meeting stile - top view - cuts / layout: cm; stock / hardware: imperial



1 T1 finish screw

GRK #8 × 1-1/2 in; T10.
Drawing length: 1.55 in.
Geometric reach: 0.845 in.
Axis: 3.75 cm from leaf edge.
Pilot/countersink: test in your actual thin trim first.

2 P2 screw before trim

Broad bearing head; set flush.
Do not crush the plywood.
Axis: 2.663 cm from leaf edge.
Select length for actual fixing.
Offset from the T1 screws along the strip's height.

Keep the seam free

Do not rigidly glue T1 to both plywood pieces.
The 0.32 cm gap remains behind the decorative trim.
P2 screw length is schematic.

Screw paths are offset along height; this schematic section shows both paths together.

*Trim is 0.236 in in this model; verify actual stock (not exactly 1/4 in). Screw profile is illustrative; follow the screw maker.

- 1 Hold the 0.32 cm seam with removable spacers.** Place P2 at the meeting edge, with the full strip and both seam edges supported by the 3½ in wide stile. Fasten P2 independently before fitting T1.
- 2 P2 fixing:** use exterior wood screws with heads that retain the plywood, seated flush for the trim. A measured **1¼ in** screw gives about **0.781 in** geometric reach into timber. Panel-to-frame spacing/count and bonding must suit the actual structural skin.
- 3 P4 inside:** the 1/5 in lining also fixes into the stile. A measured **1 in** screw reaches about **0.800 in** into timber. A low-profile pan/washer head helps retain a thin liner; test the actual head and do not crush the veneer.

COVER THE SEAM; KEEP ITS MOVEMENT SPACE

Remove the spacers once the strip is held. Leave the panel joint open beneath the trim. Keep any trim adhesive on the infill side; do not make a rigid bridge between the two plywood sheets. [2]

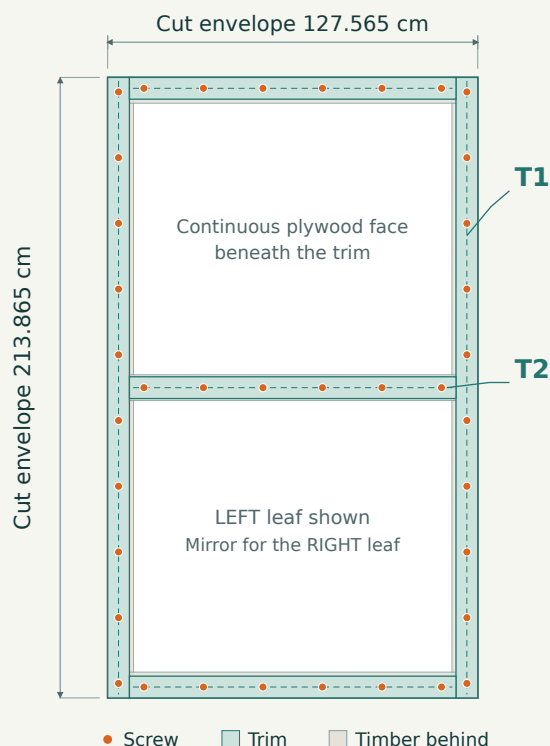
Screw on the decorative trim

Proposed retention for flat exterior plywood trim; modeled stock thickness 0.236 in.

GARAGE DOORS / ASSEMBLY DETAIL

Where to screw the decorative trim

Cuts / layout: cm - stock: 0.236 in exterior plywood trim in this model; verify actual stock



T1 / 2 vertical strips per leaf

Cut 213.865 × 7.5 cm; stock 0.236 in thick

10 screws per strip



5 cm end margin + 9 equal gaps of 22.652 cm + 5 cm
Screw row: 3.75 cm from either long edge.

T2 / 3 horizontal strips per leaf

Cut 112.565 × 7.5 cm; stock 0.236 in thick

6 screws per strip



5 cm end margin + 5 equal gaps of 20.513 cm + 5 cm
Screw row: 3.75 cm from either long edge.

38 screws per leaf

$2 \times 10 + 3 \times 6 = 38$ | 76 for both leaves

DECORATIVE TRIM ONLY

This is not the frame, hinge or plywood fixing schedule. Every trim fixing must enter sound timber underneath. Center cover C1 and any extra hardware are excluded.

Proposed screw layout for lightweight, flat trim. Confirm the actual materials and fixing product before cutting or drilling.
All orange dots land over modeled timber. Cut/layout values are not workshop tolerances. Model 0.236 in trim is not exactly 1/4 in.

Example: GRK FIN/Trim, pack label **#8 × 1½ in**, T10 drive, appropriate exterior coating or stainless variant for the wood and exposure. Drawing length: **1.55 in**; geometric reach through 0.23622 in trim + 15/32 in plywood: **0.845 in**. Measure the purchased screw. [6, 7]

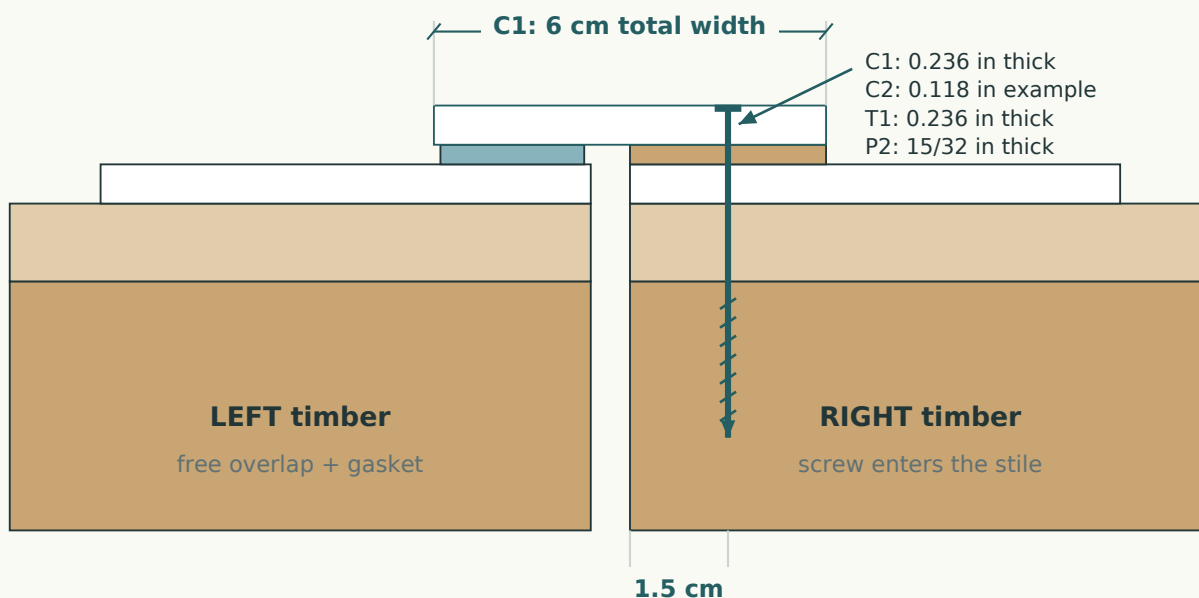
- 1 Dry-fit and clamp.** Mark the hidden stile/rail lines on the skin. Put one screw row on the strip centerline, **3.75 cm from its long edge**. Offset holes along the height to miss existing skin fasteners.
- 2 Test the drilling on matching offcuts.** A clearance hole is in the strip; a smaller pilot is in the timber. Choose them for the actual screw and wood. The small trim head is only about **0.197 in wide**: keep adequate head bearing; do not make a large generic countersink.
- 3 Drive gently to flush.** Use a low drill clutch; finish slowly. The head must retain the 0.236 in strip without crushing or pulling through it. Fill individual cosmetic recesses with exterior filler, sand and finish. No full-width glue bed is required for this proposal.

Counts for both doors: $4 \text{ T1} \times 10 + 6 \text{ T2} \times 6 = 76 \text{ trim screws}$; center-cover screws are additional. This proposed pattern holds decorative trim; it is not the frame, skin or hinge fastening schedule.

Fasten the center cover

The center cover and spacer attach to the RIGHT leaf and release with it.

TOP VIEW • OUTSIDE IS ABOVE



Cover width: 2.4 cm free LEFT overlap + 0.6 cm center gap + 3 cm RIGHT support. Modeled gasket profile: approximately 0.866 in wide within the overlap. Its compressed gap sets the actual spacer thickness.

- 1 **Close and secure LEFT; align RIGHT.** Fit the gasket first and select C2 thickness. Cut the cover and spacer ends to the actual head and bottom seals. Their 213.865 cm reference length may need end trimming.
- 2 **Mark the fixing row on RIGHT only: 1.5 cm inside its meeting edge**, centered over the 3 cm spacer. On the cover this is **4.5 cm from its left edge**. Stagger holes to miss P2/T1 screws.
- 3 **Example cover fastener:** exterior-compatible **GRK FIN/Trim #8 × 2 in**, drawing length **2.00 in**. Through the example 1.05930 in stack, geometric timber reach is **0.941 in**. Test the head and drilling as on page 23. [6, 7]
- 4 **Example cover count:** 10 screws, about 5 cm from fitted ends and evenly spaced at no more than about 25 cm. Keep the cover removable. Open RIGHT first and verify that its free overlap clears LEFT.

KEEP THE TWO LEAVES INDEPENDENT

These are weather-cover fixings. Fit handles, latches and other loaded hardware to suitable framing. No cover screw goes into the center air gap or crosses into LEFT.

Mark every frame notch

Keep sound existing joints. This cut map is for replacement F1 uprights and F2 rails.

WORKSHOP DETAIL 01 • BOTH DOORS • CUTS / MARKS: cm • STOCK: IMPERIAL

Exactly where to cut the frame

Half-lap joints: remove half the thickness from opposite faces, then overlap.

Keep sound existing joints. Replacement stock: nominal 2×4, actual 1-1/2 × 3-1/2 in.

F1 UPRIGHT ×4

CUT LENGTH 213.865 cm

OUTSIDE broad face facing you • bottom at left; top at right

HATCHED = WASTE

REMOVE HALF DEPTH

0.000

8.890

102.488

111.378

204.975

213.865

Notch depth = half the actual board thickness. For 1-1/2 in timber: cut 1.905 cm deep.

F2 CROSS RAIL ×6

CUT LENGTH 127.565 cm

INSIDE broad face facing you • a half-lap at each end

FULL-WIDTH RAIL — THE ENDS OVERLAP THE UPRIGHTS

0.000

8.890

118.675

127.565

SECTION THROUGH THE LAP

Together, the two retained halves fill 1-1/2 in.

F2 retains OUTSIDE half

F1 retains INSIDE half

Each retained half = 3/4 in for this model.

EXACT LAYOUT MARKS

F1: 0–8.890 | 102.4875–111.3775 | 204.975–213.865

F2: 0–8.890 | 118.675–127.565

Measure from one squared end; mark with the actual mating board.

The F2 rail is 127.565 cm long, including both overlaps.

Do not use the clear opening between the uprights as cut length.

Cut maps are model dimensions, not site measurements. Verify timber, leaf size and squareness before cutting.

Cuts / layout marks: cm • Purchased stock: imperial • Rounded labels are not a cutting tolerance

Member / face to remove	Notch ranges, cm, from one squared reference end
F1 upright / OUTSIDE half	0–8.89; 102.4875–111.3775; 204.975–213.865
F2 rail / INSIDE half	0–8.89; 118.675–127.565

1

Crosscut blanks square: F1 = **213.865 cm**, four total; F2 = **127.565 cm**, six total. Keep the 2×4 laid flat: **3½ in face × 1½ in depth**. No lengthwise ripping of the frame stock is planned.

2

Label faces and one reference end. Use the actual mating board to mark each notch width. Set the cut depth to half the actual thickness: **1.905 cm** for actual 1½ in thick stock. The two members lose opposite halves.

Do not cut F2 to 109.785 cm. That is the clear span between uprights. The full-width rail includes its two end overlaps. Model decimals are layout references, not a promise of site precision.

Both completed faces should be flush; combined joint thickness equals the original timber depth.

25

Cut and fit the half-lap

Start with two offcuts of the actual timber. A handsaw and sharp chisel are enough for the trial.

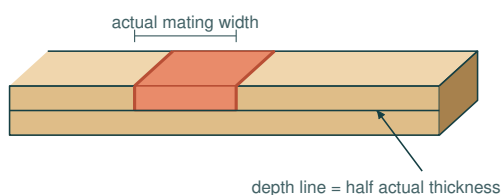
WORKSHOP DETAIL 02 • HANDSAW + SHARP CHISEL

Cut one test joint before cutting all pieces

Use two offcuts from your actual timber. Check face labels every time you turn a board.

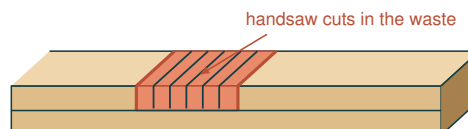
Clamp the board to a solid bench. Support the long length. Keep hands clear of the cutting path.

1 MARK THE SHOULDERS + HALF DEPTH



Square the blank to length; label OUTSIDE and INSIDE. Use the mating board to mark the lap width. Knife the shoulders and mark half depth on both narrow edges.

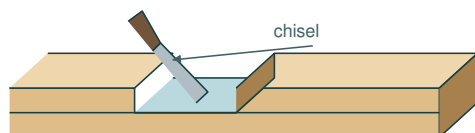
2 SAW THE SHOULDERS, THEN RELIEF CUTS



STOP short of the depth line; pare the last small amount.

Make a shallow starter on the waste side of the line. Middle notch: two shoulders. End lap: one shoulder. Add relief cuts; keep them above the depth line.

3 REMOVE WASTE; PARE THE FLOOR FLAT



Blue floor = the retained half. Finish exactly to the line.

Remove small chips between the relief cuts. Work from each edge toward the middle; take thin cuts. Keep both hands behind the sharp edge. Pare flat.

4 DRY-FIT THE OPPOSITE HALVES



Broad faces flush; shoulders touching; no rocking. If tight, pare high spots; do not hammer the joint. If loose or overcut, remake it; do not fill for fit. Dry-fit the frame flat and check equal diagonals.

*For the nominal model. Match the real board thickness. Cutting diagrams do not specify structural joint fastening.

Method reference: Lee Valley Tools, "Mastering Handsaw Cuts" (Christopher Schwarz). Illustrations are original.

Cuts / marks: cm • Stock / equipment: imperial • Process diagrams are enlarged, not full-size templates

- 1 Knife the shoulder lines and mark the depth on both edges.** Secure the board to a firm bench and support its length. Put the saw on the waste side of the line; a small chiseled starter helps locate the cut. [8]
- 2 Saw shoulders, then relief cuts.** A center notch has two shoulders; an end lap has one. Stop short of the final depth line so the last thin amount can be pared accurately.
- 3 Remove small chips with the chisel.** Work from each edge toward the middle, keeping hands behind the cutting edge. Pare the floor flat to the line. Do not lever against the retained half.
- 4 Dry-fit the complementary pieces.** Faces should be flush, shoulders touching and the joint free of rocking. Pare tight high spots; remake a loose or overcut piece. Clamp the full frame flat and compare both diagonals: reference **249.020 cm**.

AFTER THE DRY FIT

Cutting accuracy does not select the structural joint connection. Complete the confirmed bonding, fasteners and racking reinforcement for the actual timber and leaf load before adding skins or hanging.

Design notes and references

The drawings show the current V4 geometry and your stated material thicknesses. Product instructions control product-specific assembly.

DESIGN BASIS AND USE

Reference opening: **257 × 216 cm**; leaf cuts: **127.565 × 213.865 cm**. Stock: 15/32 in outside plywood, 1½ in core, 1/5 in lining; trim model 0.236 in. Use imperial for purchases and cm for cuts/marks. Quantities cover both leaves. Verify actual stock and site dimensions.

Keep sound existing framing. Pages 21–26 add workshop methods and a proposed decorative-trim fixing layout. They do not establish structural, wind, fire or thermal performance. Verify actual frame connections, panel bracing, hardware and foam covering for the installation.

[1] APA — Plywood exposure classifications — Verify the exterior-use marking; thickness alone does not define exposure suitability.

[Open primary reference](#)

[2] APA — Prevent Buckling with Proper Spacing — General panel spacing guidance; the supported 0.32 cm seam is this project's detail.

[Open primary reference](#)

[3] Henkel — LOCTITE PL 300 technical data — Check exact foam, facing, substrate, retention and cure; no sandwich-panel rating is implied.

[Open primary reference](#)

[4] Simpson Door Company — Care and finishing — Protect faces, edges and later cuts using the selected coating system.

[Open primary reference](#)

[5] Titebond III — Manufacturer limitations — Exterior-use glue is not a structural door-joint specification; stated limitations apply.

[Open primary reference](#)

[6] GRK — Finish Trim Head product — Exterior-trim application, finishes, T10 drive and pack labels. Our spacing/counts are a proposal.

[Open primary reference](#)

[7] GRK — Finish screw drawing, dated 11/01/2023 — Nominal #8 × 1½ in product: drawing L = 1.55 in; #8 × 2 in product: L = 2.00 in. Measure purchased parts.

[Open primary reference](#)

[8] Lee Valley Tools — Mastering Handsaw Cuts — Christopher Schwarz: knife-marked shoulders, waste-side saw starts and securely clamped work.

[Open primary reference](#)

Unit convention updated 09 October 2026. CAD geometry and workshop illustrations are original. Retain the exact cm cut list and 3D model with this manual. Screw reach is geometric penetration, including the tip; it is not a load-capacity calculation.